MARKING SCHEME

Model Paper — 03

Mathematics II

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Note (for teachers):

- Candidates answer **five** questions from Part A and **five** questions from Part B. Each question carries 10 marks (total 100 marks).
- The marks for each step are shown in the right-hand column; the step marks add up to the marks printed for each part of the question paper.
- Give full marks for correct alternative methods. For values read from graphs or measured from constructions, accept the ranges given. Table values may differ slightly in the last figure.

Part A — Marking Scheme

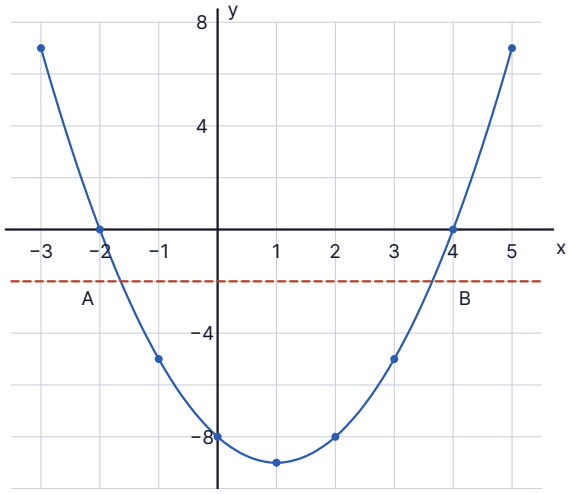
Question 1

(a)	
Interest = 82 800 − 72 000 = Rs 10 800	**1**
Rate = $\frac{10\,800}{72\,000} \times 100$	**1**
= **15%** per annum	**1**
(b)	
Total paid = 12 × 6 520 = Rs 78 240	**1**
Interest = 78 240 − 72 000 = **Rs 6 240**	**1**
(c)	
Capital repaid each month = 72 000 ÷ 12 = Rs 6 000	**1**
Amount owed during the 12 months: 72 000, 66 000, 60 000, … , 6 000; sum = 6 000 × (12 + 11 + … + 1) = 6 000 × 78 = Rs 468 000 (per month)	**1**
Interest = 468 000 × $\frac{r}{100} \times \frac{1}{12}$ = 6 240	**1**
$r = \frac{6\,240 \times 12 \times 100}{468\,000}$ = **16%** per annum	**1**
(d)	
16% > 15%, so the annual interest rate under Supun's method is higher than that of Hasini's loan.	**1**

Total marks: 10

Question 2

(i)	
$y = -5$ (symmetric about $x = 1$; the value at $x = -1$)	**1**
(ii)	
Suitable scales on both axes, axes labelled (e.g. 2 cm for 1 unit on x, 2 cm for 4 units on y)	**1**
All 9 points plotted correctly: (−3, 7), (−2, 0), (−1, −5), (0, −8), (1, −9), (2, −8), (3, −5), (4, 0), (5, 7)	**1**
Smooth parabola, minimum at (1, −9), passing through all points	**1**
(iii)	
(a) $x = -2$ and $x = 4$	**1**
(b) $y = (x + 2)(x - 4)$ $(p = 2, q = -4)$	**1**
(iv)	
(a) Line $y = -2$ drawn and read off: A ≈ (−1.6, −2), B ≈ (3.6, −2)	**1**
$-1.6 < x < 3.6$ (accept −1.7 to −1.5 and 3.5 to 3.7)	**1**
(b) AB = 3.6 − (−1.6) (using the values read)	**1**
AB ≈ **5.3 cm** (accept 5.2 – 5.4; exact value $2\sqrt{7} = 5.29$)	**1**



Total marks: 10

Question 3

(i)	
In right-angled triangle ABC (angle at B): BC = AC sin 18°20′ = 36 × sin 18°20′	**1**
sin 18°20′ = 0.3145 (tables)	**1**
BC = 36 × 0.3145 = 11.322 ≈ **11.32 m**	**1**
(ii)	
BD = 2 × BC = 2 × 11.32 = 22.64 m (C is the mid-point of BD)	**1**
tan ∠DEB = $\frac{BD}{BE} = \frac{22.64}{36}$	**1**
= 0.6289	**1**
∠DEB = **32°10′** (from tan tables)	**1**
(iii)	
Inclination of DE − inclination of AC = 32°10′ − 18°20′	**2**
= 13°50′ > 13°; so the required result is true	**1**

Total marks: 10

Question 4

(i)	
Area of triangle = $\frac{1}{2}(2x + 4)^2 = 2x^2 + 8x + 8$	**1**
Area of rectangle = $(x + 5)(x - 3) = x^2 + 2x - 15$	**1**
$(2x^2 + 8x + 8) - (x^2 + 2x - 15) = 77$	**1**
$x^2 + 6x + 23 = 77 \Rightarrow x^2 + 6x - 54 = 0$	**1**
(ii)	
$x = \frac{-6 \pm \sqrt{(36 + 216)}}{2} = \frac{-6 \pm \sqrt{252}}{2} = -3 \pm 3\sqrt{7}$	**1**
Positive value: $x = -3 + 3 \times 2.65 = 4.95$ (the negative root is rejected)	**1**
$x \approx$ **5** cm	**1**
(iii)	
Equal sides = 2 × 5 + 4 = 14 cm	**1**
Longest edge (hypotenuse) = $14\sqrt{2} = 14 \times 1.41$	**1**
= 19.74 cm > 19 cm	**1**

Total marks: 10

Question 5

(i)	
$x + y = 60$	**1**
$90x + 160y = 7\,150$ $(9x + 16y = 715)$	**1**
Eliminating x: $9x + 9y = 540$; subtract: $7y = 175$	**1**
$y = 25$ large packs	**1**
$x = 60 - 25 = 35$ small packs	**1**
(ii)	
Small pack contains n lanterns, large pack contains $n + 4$	**1**
$35n + 25(n + 4) = 1\,000$	**2**
$60n = 900$	**1**
$n =$ **15** lanterns	**1**

Total marks: 10

Question 6

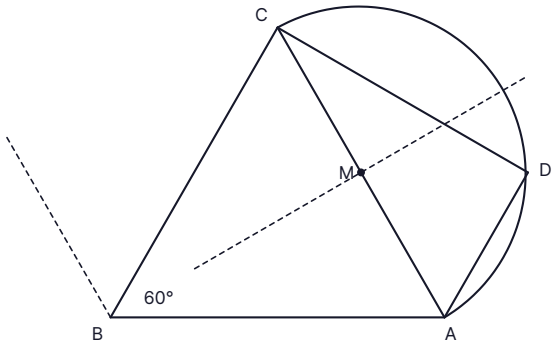
(i)	
Mid-values 35.5, 45.5, 55.5, 65.5, 75.5; $\Sigma f = 40$	**1**
$\Sigma fx = 142 + 364 + 666 + 655 + 453 = 2\,280$	**1**
Mean $= 2\,280 \div 40 =$ **57 kg** per day	**1**
(ii)	
Expected total $= 57 \times 30$	**1**
= **1 710 kg**	**1**
(iii)	
Income $= 1\,710 \times 48$	**2**
= **Rs 82 080**	**1**
(iv)	
Days with less than 51 kg: classes 31 – 40 (4 days) and 41 – 50 (8 days); minimum masses 31 kg and 41 kg	**1**
$4 \times 31 + 8 \times 41 = 124 + 328 =$ **452 kg**	**1**

Total marks: 10

Part B — Marking Scheme

Question 7

(i)	
AB = 8 cm drawn (7.9 – 8.1 accepted)	**1**
120° at B constructed with two 60° arcs of the same radius (arcs visible), ∠ABX = 120° (118° – 122° accepted)	**1**
Bisector BY constructed with arcs visible, ∠ABY = 60°	**1**
(ii)	
C marked on BY with BC = 8 cm (compass arc from B), AC joined; triangle ABC is equilateral, AC = 8 cm (7.9 – 8.1)	**1**
(iii)	
Perpendicular bisector of AC drawn with arcs of equal radius from A and C visible	**1**
Mid-point M marked on AC	**1**
Semicircle on AC as diameter (radius 4 cm, centre M) drawn on the side of AC away from B	**1**
(iv)	
Line through A parallel to BC constructed (angle 60° copied at A, arcs visible)	**1**
D marked where it meets the semicircle (AD ≈ 4 cm, 3.9 – 4.1), CD joined (CD ≈ 6.9 cm, 6.8 – 7.0)	**1**
(v)	
∠ADC = 90° (angle in a semicircle); ∠DAC = ∠ACB = 60° (alternate angles, AD ∥ BC); so ∠ACD = 180° − 90° − 60° = **30°**	**1**



Total marks: 10

Question 8

(i)	
In triangles BFE and CGE: $\angle BEF = \angle CEG$ (vertically opposite angles)	**1**
$\angle FBE = \angle GCE$ (alternate angles, AB ∥ DG)	**1**
BE = EC (E is the mid-point of BC) $\therefore \triangle BFE \cong \triangle CGE$ (angle, side, angle)	**1**
(ii)	
BF = CG (corresponding sides of congruent triangles)	**1**
BF ∥ CG (AB ∥ DC). One pair of opposite sides equal and parallel, so BFCG is a parallelogram (alternatively: the diagonals BC and FG bisect each other at E, since FE = EG)	**1**
(iii)	
In triangles AFH and BFE: $\angle AFH = \angle BFE$ (vertically opposite angles)	**1**
$\angle FAH = \angle FBE$ (alternate angles, AH ∥ BE since AD ∥ BC); hence $\angle AHF = \angle BEF$ also; the triangles are equiangular	**1**
(iv)	
From similar triangles: $\frac{AH}{BE} = \frac{AF}{FB}$	**1**
$AF = \frac{2}{5}AB \Rightarrow FB = \frac{3}{5}AB$, so $\frac{AF}{FB} = \frac{2}{3} \Rightarrow AH = \frac{2}{3}BE$	**1**
$BE = \frac{1}{2}BC = \frac{1}{2}AD$, so $AH = \frac{2}{3} \times \frac{1}{2}AD = \frac{1}{3}AD$	**1**

Total marks: 10

Question 9

(i)	
Volume of a pyramid = $\frac{1}{3} \times (2a)^2 \times 3a = 4a^3$; of 6 pyramids = $24a^3$	**1**
Volume of cylinder = $\pi r^2 \times 4r = 4\pi r^3$	**1**
$24a^3 = 4\pi r^3 \Rightarrow a^3 = \frac{\pi}{6} r^3$	**1**
(ii)	
$a^3 = \frac{3.14 \times 6.3^3}{6}$; lg a^3 = lg 3.14 + 3 lg 6.3 − lg 6	**1**
= 0.4969 + 3 × 0.7993 − 0.7782 = 0.4969 + 2.3979 − 0.7782 = 2.1166 $\Rightarrow$ $a^3 \approx 130.8$ (130 – 131)	**1**
lg a = 2.1166 ÷ 3 = 0.7055	**1**
a = 5.07 ≈ **5 cm**	**1**
(iii)	
With a = 5: cuboid 10 × 10 × 15 = 1 500 cm^3	**1**
Pyramid = $\frac{1}{3} \times 100 \times 15$ = 500 cm^3	**1**
Remaining volume = 1 500 − 500 = **1 000 cm^3**	**1**

Total marks: 10

Question 10

(i)	
11, 15, 19	**1**
(ii)	
$T_{13} = 11 + 12 \times 4$	**1**
= **59** skips	**1**
(iii)	
$S_{13} = \frac{13}{2}(11 + 59)$	**1**
= **455** skips	**1**
(iv)	
Ishara: $S_{14} = \frac{14}{2}[2a + 13 \times 3] = 455$	**1**
$2a + 39 = 65$	**1**
a = **13** skips	**1**
(v)	
$11 + 4(n - 1) = 13 + 3(n - 1) \Rightarrow n - 1 = 2$	**1**
$n = 3$: the **3rd day** (both skip 19 times)	**1**

Total marks: 10

Question 11

(i)	
$\angle OAP = \angle OBP = 90°$ (a tangent is perpendicular to the radius at the point of contact)	**1**
$\angle OAP + \angle OBP = 180°$ (opposite angles); therefore PAOB is a cyclic quadrilateral	**1**
$\angle AOB = 2\angle ACB$ (angle at the centre is twice the angle at the circumference, standing on the same arc AB)	**1**
$\angle APB + \angle AOB = 180°$ (opposite angles of the cyclic quadrilateral PAOB)	**1**
Therefore $\angle APB + 2\angle ACB = 180°$	**1**
(ii)	
$\angle PBA$ is equal to $\angle PAB$: naming	**1**
PA = PB (tangents drawn from an external point are equal), so $\triangle PAB$ is isosceles and the base angles are equal	**1**
$\angle ACB$ is equal to $\angle PAB$: naming (C on the major arc, in the alternate segment)	**1**
Angle between the tangent PA and the chord AB = angle in the alternate segment (tangent–chord theorem)	**2**

Total marks: 10

Question 12

(i)	
Volunteers who helped with at least one task = 90 − 8 = 82	**1**
Exactly one task = 82 − 36	**1**
= **46**	**1**
(ii)	
Let the number who helped with all three = x; then exactly two tasks = x	**1**
$x + x = 36$ (at least two tasks)	**1**
x = **18**	**1**
(iii)	
Let D and C only = y, F and C only = $2y$; $6 + y + 2y = 18$ (exactly two tasks)	**1**
$3y = 12$; y = **4**	**1**
(iv)	
n(F) = F only + (D, F only) + (F, C only) + all three: 40 = F only + 6 + 8 + 18	**1**
F only = **8**	**1**

Region	D only	F only	C only	D, F only	D, C only	F, C only	D, F, C	None
Number	d	8	c	6	4	8	18	8

(The numbers who helped with decorations only and cleaning only are not fixed by the data; they satisfy $d + c = 38$.)

Total marks: 10